

DISEASES

Std. 6-7 | Homi Bhabha Examination

These notes include all the points visible in the 3 pages/images you provided, reorganised for easier learning. I have also added a few Homi Bhabha-oriented concepts, connections, exceptions and exam traps beyond the textbook.

1. What is a Disease?

A disease is a condition of the body or a part of the body in which its normal functions are disturbed.

A disease may involve:

- Structural / morphological disturbance – a change in the structure of a body part.
- Physiological / functional disturbance – a disturbance in the normal functioning of the body.
- Psychological disturbance – disturbance affecting mental or psychological functions.

Diseases may be caused by external agencies or by internal factors such as deficiency of nutrients.

External agencies may be:

- Parasitic – caused by disease-causing organisms living on or inside the host.
- Non-parasitic – caused by factors other than parasites, such as nutritional deficiencies.

Simple definition to remember: **Disease = A condition in which the normal functioning of the body is disturbed.**

2. Causes of Diseases

The major causes mentioned in the lesson are:

1. Pathogens

Pathogens are disease-causing organisms.

Important pathogens include:

- Bacteria
- Viruses
- Fungi
- Protozoans
- Worms
- Mites

2. Lack of a Healthy Diet

A diet lacking essential nutrients can make a person unhealthy.

For example:

- Lack of Vitamin C → Scurvy
- Lack of Vitamin D → Rickets
- Lack of iron → Anaemia

These are generally classified as deficiency diseases.

3. Lack of Public Facilities

Poor public facilities increase the chances of diseases.

Important facilities include:

- Clean drinking water
- Proper sewage disposal
- Proper garbage disposal
- Prevention of contamination of food and water

Poor sanitation and contaminated water/food can allow infectious diseases to spread.

3. Classification of Diseases

A. Congenital Diseases

Congenital diseases are present from birth.

Examples:

- Haemophilia
- Colour blindness
- Sickle-cell anaemia
- Down's syndrome
- Albinism

Remember: Congenital → Present from birth

B. Acquired Diseases

Diseases acquired by an organism after birth are called acquired diseases.

According to the provided textbook, acquired diseases are not inherited, i.e. they do not pass from one generation to another.

Acquired diseases are broadly classified into:

- I. Communicable / Infectious Diseases
- II. Non-communicable / Non-infectious Diseases

4. Communicable / Infectious Diseases

These diseases are caused by pathogens / infectious agents.

They can spread from an infected person to a healthy person.

Examples include:

- Tuberculosis
- Malaria
- Diarrhoea
- Chickenpox
- Influenza

Major ways of spread

Touch | Food | Water | Air | Vector
Other specific routes include: Soil through injuries | Animal bites

5. Non-Communicable / Non-Infectious Diseases

These diseases cannot spread from an infected person to a healthy person.

The lesson gives four important categories:

A. Degenerative Diseases

These involve deterioration or degeneration of body organs/tissues.

Examples:

- Heart diseases
- Nephritis

B. Deficiency Diseases

These occur due to deficiency of essential nutrients.

Examples:

- Scurvy
- Rickets
- Anaemia

C. Allergies

An allergy is an abnormal reaction of the body to certain substances.

Example: Hay fever

D. Cancer

Cancer is a non-communicable disease involving uncontrolled growth of abnormal cells.

6. Infectious Agents / Pathogens

The major infectious agents in the lesson are:

- 1. Bacteria
- 2. Viruses
- 3. Protozoans
- 4. Helminths / Worms
- 5. Fungi

The table in the textbook also mentions mites as the cause of scabies.

7. Bacteria

Bacteria are:

- Unicellular
- Prokaryotic
- Microscopic
- Capable of reproducing very quickly

Diseases caused by bacteria

Disease	Pathogen
Typhoid	Bacteria
Cholera	Bacteria
Tuberculosis	Bacteria
Anthrax	Bacteria
Diphtheria	Bacteria
Tetanus	Bacteria
Pneumonia	Bacteria
Syphilis	Bacteria
Gonorrhoea	Bacteria
Leprosy	Bacteria

□ Homi Bhabha memory group

Bacteria → Typhoid, Cholera, TB, Anthrax, Tetanus, Diphtheria

8. Viruses

The textbook describes viruses as a connecting link between living and non-living organisms.

Viruses reproduce only inside suitable living cells.

Diseases caused by viruses

- Common cold
- Influenza
- Measles
- Mumps
- Poliomyelitis / Polio
- Rabies
- Smallpox
- Chickenpox
- Yellow fever
- AIDS
- Dengue fever

□ Important

- Rabies → Virus
- Measles → Virus
- Polio → Virus
- Chickenpox → Virus
- Dengue → Virus

9. Protozoans

Protozoans are:

- Unicellular
- Eukaryotic organisms

They can cause several diseases.

Important examples

Disease	Causative organism
Malaria	Plasmodium
Kala-azar / Black fever	Leishmania
Amoebic dysentery	Protozoan
Sleeping sickness	Protozoan

□ Very Important

- Malaria → Plasmodium
- Kala-azar / Black fever → Leishmania

10. Helminths / Worms

Helminths are multicellular worms, many of which are found in the intestine.

Important diseases include:

Taeniasis

Caused by tapeworm.

Ascariasis

Caused by roundworm.

Filariasis

Caused by filaria worm.

Disease	Worm
Taeniasis	Tapeworm
Ascariasis	Roundworm
Filariasis	Filaria worm

Exam Trap

**Taeniasis → Tapeworm
 Ascariasis → Roundworm
 Do not interchange them.**

11. Fungi

Fungi mentioned in the lesson are:

- Multicellular
- Eukaryotic
- Heterotrophic

They can cause skin infections.

Important diseases

- Ringworm
- Athlete's foot

Important Homi Bhabha Trap

**Despite its name, ringworm is NOT caused by a worm.
 Ringworm → Fungus**

12. Mites

Scabies is caused by mites.

**Remember:
 Scabies → Mite**

13. Complete Pathogen-Disease Chart

This table combines the disease list from your textbook pages into one revision chart.

Infectious agent	Diseases
Viruses	Common cold, Influenza, Measles, Mumps, Polio, Rabies, Smallpox, Chickenpox, Yellow fever, AIDS, Dengue fever
Bacteria	Cholera, Typhoid, Tuberculosis, Anthrax, Diphtheria, Tetanus, Pneumonia, Syphilis, Gonorrhoea, Leprosy
Protozoans	Malaria, Amoebic dysentery, Sleeping sickness, Kala-azar
Fungi	Ringworm, Athlete's foot
Worms	Filariasis, Ascariasis, Taeniasis
Mites	Scabies

14. Important Disease → Exact Causative Organism

For Homi Bhabha, learn these exact pairs.

Disease	Causative organism / agent	Group
Malaria	Plasmodium	Protozoan
Kala-azar / Black fever	Leishmania	Protozoan
Taeniasis	Tapeworm	Helminth

Disease	Causative organism / agent	Group
Ascariasis	Roundworm	Helminth
Filariasis	Filaria worm	Helminth
Scabies	Mite	
Ringworm	Fungus	

15. Spread of Diseases

Infectious diseases are called communicable diseases because they can spread from an infected person to a healthy person.

The method of transmission depends upon the particular pathogen.

Major routes include:

- 1. Air
- 2. Water
- 3. Food
- 4. Physical contact / Touch
- 5. Vectors
- 6. Soil through injuries
- 7. Animal bites

16. Direct Transmission

In direct transmission, the pathogen is transmitted from an infected person/source to a healthy person directly, without an intermediate agent.

The textbook gives three important examples.

A. Contact with an Infected Person

Some diseases spread through actual physical contact between an infected person and a healthy person.

Examples:

- Chickenpox
- Smallpox
- Ringworm

Such diseases are called contagious diseases.

Remember: Physical contact → Contagious disease

17. Contact with Soil

The infectious agent responsible for tetanus can enter the human body through an injury contaminated with soil.

Example

A person gets a cut/injury while working with soil.

If the wound becomes contaminated with the organism causing tetanus, infection may occur.

Remember: Contaminated soil + wound → Tetanus

18. Animal Bites

The rabies virus can be transmitted to humans through the bite of an infected/rabid animal.

Examples mentioned in the textbook:

- Rabid dog
- Monkey

Remember: Animal bite → Rabies

19. Direct vs Indirect Transmission

This distinction is useful for Homi Bhabha questions.

Direct Transmission	Indirect Transmission
Pathogen reaches the person without an intermediate agent.	An intermediate medium or agent helps transmit the pathogen.
Direct contact	Air/droplets
Soil entering an injury	Contaminated food
Animal bite	Contaminated water
	Vectors such as insects

20. Vectors

A vector is an organism that carries a pathogen from one host to another.

Common example

Mosquito → Vector

Mosquitoes can transmit diseases such as malaria and dengue.

Important distinction

Plasmodium causes malaria.

Mosquito acts as a vector for transmission of malaria.

So:

Causative organism ≠ Vector

This distinction is highly useful in MCQs.

21. Air-Borne / Droplet Transmission

Some infectious diseases can spread through droplets released when an infected person:

- Coughs
- Sneezes
- Talks

The textbook specifically refers to the air/droplet method.

Examples from the disease list include:

- Common cold
- Influenza

22. Food and Water Transmission

Contaminated food and water can transmit infectious organisms.

Poor sanitation and improper disposal of sewage can increase contamination.

Examples associated with food/water transmission include diseases such as:

- Cholera
- Typhoid
- Diarrhoeal diseases

□ Prevention

- Drink safe water.

- Wash food properly.
- Maintain sanitation.
- Dispose of sewage properly.
- Keep food covered.

23. Important Differences

Congenital vs Acquired

Congenital	Acquired
Present from birth	Develop/acquired after birth
Examples: haemophilia, colour blindness	Examples: malaria, tuberculosis
May be genetic or developmental	Usually arises after birth

Communicable vs Non-Communicable

Communicable	Non-Communicable
Can spread between people	Does not spread person-to-person
Usually involves infectious agents	May result from deficiency, degeneration, allergy etc.
Malaria, TB, measles	Scurvy, rickets, cancer

Bacteria vs Protozoa

Bacteria	Protozoa
Unicellular	Unicellular
Prokaryotic	Eukaryotic
Microscopic	Microscopic
Example: bacteria causing TB	Plasmodium causing malaria

□ **Key difference**
Bacteria → Prokaryotic
Protozoa → Eukaryotic

24. High-Yield Homi Bhabha Facts

Disease Classification

Disease ↓ **Congenital / Acquired** ↓ **Acquired** → **Communicable / Non-communicable**

Infectious Agents

Bacteria → Typhoid, Cholera, TB, Tetanus

Viruses → Measles, Polio, Rabies, Chickenpox

Protozoa → Malaria, Kala-azar

Worms → Taeniasis, Ascariasis, Filariasis

Fungi → Ringworm, Athlete's foot

Mites → Scabies

25. □ Homi Bhabha One-Liners

Question	Answer
1. Disease present from birth?	Congenital disease

Question	Answer
2. Disease acquired after birth?	Acquired disease
3. Disease that can spread from person to person?	Communicable disease
4. Disease that does not spread from person to person?	Non-communicable disease
5. Malaria is caused by?	Plasmodium
6. Plasmodium belongs to?	Protozoa
7. Kala-azar is caused by?	Leishmania
8. Black fever is another name for?	Kala-azar
9. Taeniasis is caused by?	Tapeworm
10. Ascariasis is caused by?	Roundworm
11. Filariasis is caused by?	Filaria worm
12. Scabies is caused by?	Mites
13. Ringworm is caused by?	Fungi
14. Ringworm is actually caused by a worm?	No
15. Tetanus can enter through?	Injuries contaminated with soil
16. Rabies can be transmitted through?	Bite of an infected animal
17. Rabies is caused by?	Virus
18. Bacteria are?	Unicellular, prokaryotic, microscopic organisms
19. Protozoans are?	Unicellular, eukaryotic organisms
20. Fungi are?	Multicellular, eukaryotic, heterotrophic organisms

26. □ Conceptual / Application Questions

Q1.

A student gets a wound while playing in soil. Which disease mentioned in the lesson can be associated with entry of an infectious agent through the wound?

Answer: Tetanus

Q2.

A child develops a disease after being bitten by a rabid dog. Is this direct or indirect transmission?

Answer: Direct transmission through an animal bite

Q3.

A disease is caused by a unicellular eukaryotic organism. Which group should you suspect?

Answer: Protozoa

Q4.

A disease is caused by a multicellular worm living in the intestine. Which group is responsible?

Answer: Helminths

Q5.

A student sees the word "ringworm" and classifies it under worms. Is this correct?

Answer: No. Ringworm is caused by a fungus.

Q6.

Two students give these answers:
Student A: Malaria → mosquito
Student B: Malaria → Plasmodium
Who has identified the causative organism correctly?

Answer: Student B
Explanation: Plasmodium is the causative protozoan; a mosquito acts as a vector in malaria transmission.

Q7.

A disease is caused by deficiency of an essential nutrient and cannot spread from one person to another. What type of disease is it?

Answer: Non-communicable deficiency disease

Q8.

A disease is present in a child from birth. Under which broad classification does it fall?

Answer: Congenital disease

27. □ Ultimate Revision Map

DISEASES

1. Congenital → Present from birth → Haemophilia, Colour blindness, Sickle-cell anaemia, Down's syndrome, Albinism
2. Acquired
A. Communicable / Infectious → Caused by pathogens → Spread by touch, food, water, air, vectors etc.
B. Non-communicable / Non-infectious → Degenerative → Deficiency → Allergies → Cancer

Pathogen Map

BACTERIA → Typhoid, Cholera, TB, Tetanus, Diphtheria
VIRUSES → Measles, Mumps, Polio, Rabies, Chickenpox, Dengue
PROTOZOA → Malaria → Plasmodium → Kala-azar → Leishmania
HELMINTHS → Taeniasis → Tapeworm → Ascariasis → Roundworm → Filariasis → Filaria worm
FUNGI → Ringworm → Athlete's foot
MITES → Scabies

□ Final Memory Chain

Congenital → Birth
Acquired → After birth
Communicable → Can spread
Non-communicable → Cannot spread person-to-person
Bacteria → Prokaryotic
Protozoa → Unicellular eukaryotic
Fungi → Multicellular eukaryotic
Malaria → Plasmodium
Black fever → Leishmania
Taeniasis → Tapeworm
Ascariasis → Roundworm
Filariasis → Filaria worm
Scabies → Mite
Ringworm → Fungus
Soil + wound → Tetanus
Animal bite → Rabies
Vector → Carries/transmits pathogen

This version retains the substantive points from all three supplied pages while reorganising them into an exam-oriented structure and adding clarification where it helps with Homi Bhabha-style MCQs and application questions.